

Darukaa.Earth Knowledge Synthesis Report 05

Pollution, Urban Expansion and Species Richness

Document type: institutional scientific synthesis

Topics: human_impact, pollution, urban land, species_richness

1. Purpose

Urban expansion and pollution reduce species richness through habitat replacement, isolation, and chemical stress. This report supports reasoning when users report high pollution, low species richness, and urban growth.

2. Habitat destruction by sealing and conversion

Urban land replaces soil and vegetation with impervious surfaces. Remaining habitat is often fragmented parks, drains, and leftover wasteland. Species that need large, quiet, or specialised habitats decline. CBD urban biodiversity work and IPBES land-use chapters treat urbanisation as a major conversion pressure.

3. Pollution pathways

Air, water, soil, and light pollution affect physiology and behaviour. Heavy metals and persistent organic pollutants can reduce soil microbial diversity and sensitive aquatic invertebrates. Nutrient pollution can simplify plant communities toward a few nitrophilous species, lowering plant diversity even if biomass is high.

High pollution plus low species richness is therefore a coupled diagnosis: chemical filters plus habitat filters. Planting ornamental species without reducing pollutant loads may fail for sensitive taxa.

4. Interventions

- Source control: reduce effluent, dust, agrochemical drift, or industrial discharge depending on the actual pollutant.
- Native vegetation buffers along waterways and road edges to intercept sediments and provide habitat.
- Habitat corridors connecting remaining patches.
- Restore unsealed soils where possible rather than only adding planters on concrete.

Impacted metrics: pollution pressure downward if sources are controlled; habitat diversity upward; species richness upward only if both habitat and toxicity improve. Time horizon: water quality and floral resources can change in 1-3 years; recovery of long-lived or area-sensitive species is longer and uncertain.

5. Evidence caution

Do not attribute a numeric species recovery to a generic tree-planting campaign. Urban ecology outcomes are highly site-specific.

6. Source basis

IPBES land-use and pollution drivers; CBD urban biodiversity guidance; WHO and environmental agency consensus that chemical pollution is a biodiversity pressure. Pages refer to this synthesis.